

LiteAT: A Data-Lightweight and User-Adaptive VR Telepresence System for Remote Education

Yuxin Shen , Wei Liang* , Jianzhu Ma 

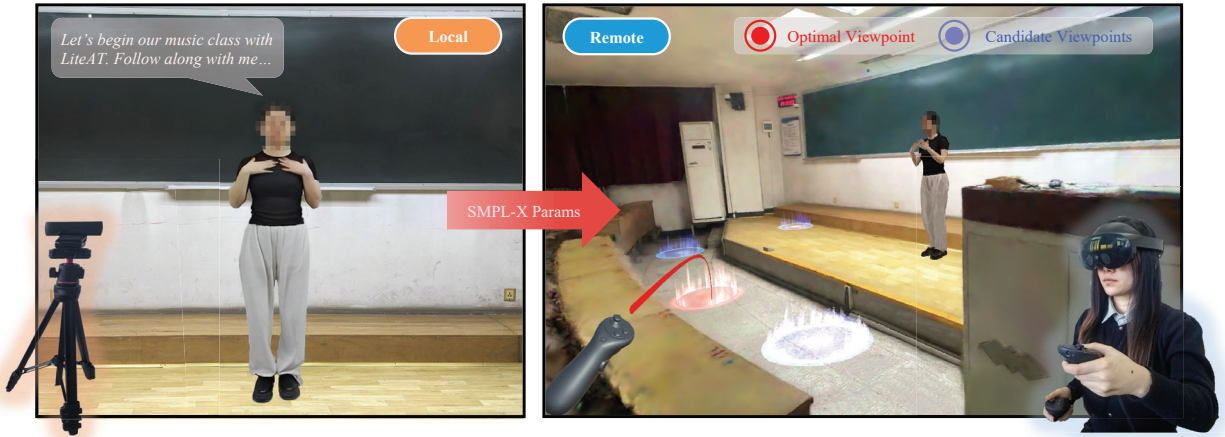


Fig. 1: LiteAT, a data-lightweight and user-adaptive VR telepresence system for remote education. (Left) A local teacher is captured using a single camera, and the estimated SMPL-X parameters are transmitted to the remote side. (Right) On the remote side, a dynamic human model driven by SMPL-X parameters and a pre-reconstructed classroom environment, both represented using 3D Gaussians, are jointly rendered in the student's VR headset. A set of high-quality, user-adaptive candidate viewpoints is generated to support efficient and personalized scene exploration, allowing the student to navigate toward the optimal viewpoint.

Abstract—In educators' ongoing pursuit of enriching remote education, Virtual Reality (VR)-based telepresence has shown significant promise due to its immersive and interactive nature. Existing approaches often rely on point cloud or NeRF-based techniques to deliver realistic representations of teachers and classrooms to remote students. However, achieving low latency is non-trivial, and maintaining high-fidelity rendering under such constraints poses an even greater challenge. This paper introduces LiteAT, a data-lightweight and user-adaptive VR telepresence system, to enable real-time, immersive learning experiences. LiteAT employs a Gaussian Splatting-based reconstruction pipeline that integrates an SMPL-X-driven dynamic human model with a static classroom, supporting lightweight data transmission and high-quality rendering. To enable efficient and personalized exploration in the virtual classroom, we propose a user-adaptive viewpoint recommendation framework that dynamically suggests high-quality viewpoints tailored to user preferences. Candidate viewpoints are evaluated based on multiple visual quality factors and are continuously optimized based on recent user behavior and scene dynamics. Quantitative experiments and user studies validate the effectiveness of LiteAT across multiple evaluation metrics. LiteAT establishes a versatile and scalable foundation for immersive telepresence, potentially supporting real-time scenarios such as procedural teaching, multimodal instruction, and collaborative learning.

Index Terms—Remote education, telepresence, gaussian splatting, viewpoint recommendation

1 INTRODUCTION

Remote education has emerged as an integral part of contemporary education, helping bridge geographical gaps between teachers and students. While 2D videoconferencing platforms (e.g., Zoom, Google

Classroom) support real-time communication, they fail to capture the richness of in-person instruction. To address this gap, telepresence has explored 3D visualization methods like volumetric [62] and holographic projections [38], enabling a sense of co-location. However, these approaches often lack a strong sense of embodiment and co-presence, hindering engagement and retention [20].

The advent of VR technology has catalyzed transformative advancements in telepresence systems, attracting significant attention across diverse domains, including diagnostics [1], training [23, 25, 56], and education [43]. VR offers unprecedented immersion, bringing students and teachers closer to real-world experiences [39], thereby enhancing comprehension and retention [8]. It also enables learners to actively engage with virtual environments and maintain visual interaction with instructors, shifting their role from passive recipients to active participants. This transition enhances spatial awareness, and fosters improved focus and cognitive engagement [40, 49].

The immersive and realistic experience of VR telepresence relies heavily on visual quality and low latency. Conventional methods use RGB-D cameras to capture point clouds [21, 52] or apply multi-view fusion to create 3D meshes [35]. Although these methods offer low-

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- Yuxin Shen is with the School of Computer Science, Beijing Institute of Technology, Beijing, China. E-mail: shenyuxin@bit.edu.cn.
- Wei Liang is with the School of Computer Science, Beijing Institute of Technology, Beijing, China, Yangtze Delta Region Academy of Beijing Institute of Technology, Jiaxing, China, and State Key Laboratory of General Artificial Intelligence, BIGAI, Beijing, China. E-mail: liangwei@bit.edu.cn.
- Jianzhu Ma is with the Department of Electronic Engineering, Tsinghua University, Beijing, China, and Institute for AI Industry Research, Tsinghua University, Beijing, China. E-mail: majianzhu@tsinghua.edu.cn.
- * indicates the corresponding author.

latency transmission, the resulting models often exhibit artifacts due to sensor noise and topological errors. To improve visual fidelity, recent studies have explored Neural Radiance Field (NeRF)-based methods [44], which employ continuous volumetric representations to produce photorealistic, view-consistent outputs with enhanced geometry and texture [31]. However, NeRFs demand long training times, making them unsuitable for latency-sensitive VR scenarios. Moreover, both pipelines involve complex encoding and decoding, increasing susceptibility to transmission delays and data loss.

These limitations pose substantial barriers to the advancement of immersive remote VR education. In global or underserved regions, unreliable or low-bandwidth networks often hinder access to high-quality VR learning environments [36]. Even when connectivity is available, insufficient real-time responsiveness can disrupt presence and diminish engagement in bidirectional interactions. Scalability remains a concern, as supporting many concurrent users can strain network infrastructure and degrade performance [51]. Furthermore, high latency and degraded visual fidelity impair learners' ability to follow instructional content, ultimately undermining comprehension and learning outcomes [53]. Collectively, these challenges underscore the urgent need for a lightweight and low-latency remote VR education solution capable of maintaining stable performance across diverse network conditions.

Gaussian Splatting, as a recent breakthrough in volume rendering, offers a compelling balance between visual fidelity and performance [19]. Its compact, explicit, and rasterization-friendly representation enables efficient optimization and real-time, high-fidelity rendering, making it particularly well-suited for dynamic, latency-sensitive telepresence. This motivates our design of **LiteAT**, a data-lightweight (**Lite**) and user-Adaptive VR Telepresence system built on a **Gaussian Splatting-based reconstruction pipeline** that incorporates a *lightweight data transmission strategy* and real-time rendering. As illustrated in Fig. 1, LiteAT captures the teacher on the local side and continuously estimates SMPL-X parameters to reflect real-time body motion. Instead of transmitting rendered images ($\sim 10^2$ KB) or dense point clouds ($\sim 10^3$ KB), LiteAT transmits lightweight SMPL-X files (~ 2 KB), significantly reducing bandwidth demands and improving responsiveness. On the remote side, these parameters drive reconstruction of a dynamic human model, rendered alongside a pre-reconstructed static classroom directly within the student's VR headset. This pipeline seamlessly integrates motion fidelity, spatial context, and rendering efficiency, delivering an immersive, responsive, high-quality classroom experience that closely approximates in-person learning.

To support flexible and personalized exploration within LiteAT, we propose a **user-adaptive viewpoint recommendation framework**, as shown in Fig. 1. This framework allows students to explore the virtual classroom through optimal viewpoints that balance visual quality and user preferences. We adopt a *graph-based modeling approach*, representing each candidate viewpoint as a node scored by training-set similarity, human proximity, environment occlusion, and historical user attention. Edges are constrained by directional continuity to ensure smooth viewpoint transitions. To enhance personalization, we also consider individual differences in exploration behavior. Studies [30, 34] show that users prioritize different viewing strategies, ranging from broader contextual perspectives to fine-grained instructional detail. Building on these insights, we introduce a *dynamic optimization mechanism* that continuously refines recommendations based on recent user behavior and scene dynamics. It consists of two components: (1) a Ridge regression model [11] that dynamically adjusts the weights of the scoring factors according to current user behavior, and (2) a Finite Difference Method (FDM) method [32] that refines the spatial positions of candidate nodes to better align with evolving user preferences.

Finally, we conduct quantitative experiments and user studies to evaluate the effectiveness of LiteAT across key metrics, including latency, visual quality, spatial awareness, social presence, and preference alignment. Beyond classroom instruction, LiteAT shows strong potential for broader applications, such as laboratory training, procedural teaching, and collaborative learning, that demand not only real-time performance and visual fidelity but also rich interaction and social presence. We envision this work as a foundation for future research toward more

general, multimodal, and multi-user telepresence systems. Our work makes three key contributions:

- We present LiteAT, a VR telepresence system that combines lightweight data transmission with high-fidelity Gaussian Splatting-based rendering, addressing the trade-off between real-time performance and visual quality in remote education.
- We propose a graph-based, user-adaptive viewpoint recommendation framework that integrates visual quality scores and dynamically adapts to user preferences and scene dynamics.
- We validate LiteAT through quantitative experiments and comparative user studies, showing improvements in latency, spatial awareness, social presence, and personalized user experience.

2 RELATED WORKS

2.1 Remote Education

Remote education, commonly delivered via platforms like Zoom and Microsoft Teams, provides flexible access for geographically dispersed learners [18]. However, 2D videoconferencing often leads to cognitive fatigue and distraction, reducing sustained engagement [6]. To address this, researchers have proposed methods to enhance student-teacher interaction, such as physical object sharing to bridge digital and tangible experiences [12], and head-pose detection to monitor attention and adapt instruction [3]. Nonetheless, the inherent limitations of 2D interfaces still hinder their ability to replicate the fluid, intuitive interactions of face-to-face communication.

To address these issues, educators have explored 3D display technologies as more effective alternatives, like autostereoscopic displays [17, 62] and holographic projections [38, 48]. For instance, Zhang et al. [62] propose a 3D videoconferencing system that generates life-sized 3D projections of participants, improving depth perception. Pejsa et al. [38] introduce Room2Room, a holographic telepresence system that uses augmented projections to embed remote users into a shared physical space, enabling more intuitive interaction. However, such systems remain constrained by physical setup and cannot fully support embodied learning experiences.

With the rise of consumer-grade and wireless VR devices (e.g., Vision Pro, Meta Quest Pro), virtual learning environments are increasingly adopted in remote education. VR immerses students in interactive and realistic spaces, promoting deeper engagement, stronger presence, and improved retention [4, 9, 24, 52]. Battipede et al. [4] assess a VR physics laboratory, demonstrating notable gains in conceptual understanding and user satisfaction. Similarly, Yu et al. [58] present a hybrid VR teaching system that enables students to interact with virtual objects and perform hands-on operations, resulting in increased engagement and participation. In response to the growing trend of remote VR education, our research explores its potential to support more embodied and immersive learning experiences.

2.2 3D Telepresence

3D telepresence aims to seamlessly integrate a remote user into a local environment via user embodiment, achieved through virtual avatars [15, 47, 50, 54] or reconstructed human representations [16, 21, 35, 44, 52]. While avatars are easy to animate using motion tracking, they lack social presence, behavioral realism, and perceived humanness [59]. In contrast, reconstructed representations preserve a user's appearance, offering a more lifelike presence. These methods rely on point cloud techniques using RGB-D cameras for 3D reconstruction. For instance, Kowalski et al. [21] introduce LiveScan3D, a system that employs Kinect V2 sensors to generate point cloud reconstructions and transmit them to the remote side. Similarly, Orts-Escolano et al. [35] propose Holoportation, a system that enables high-quality 3D telepresence by using multiple RGB-D cameras to reconstruct a fully textured, volumetric representation of users, allowing for real-time communication.

Despite these advancements, reconstruction techniques face challenges such as sensor noise, incomplete meshes, and missing data, which degrade visual fidelity and limit the accurate capture of fine

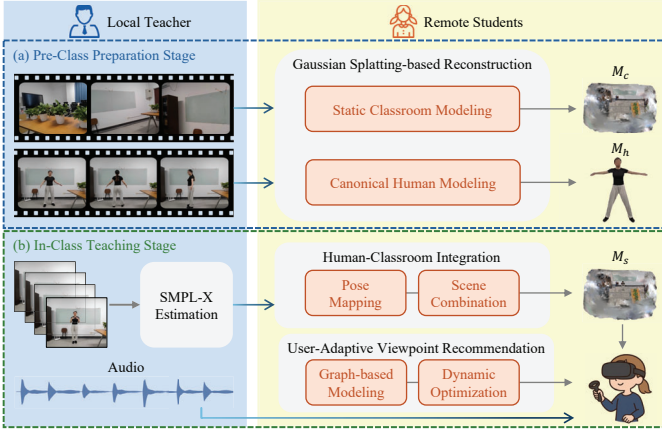


Fig. 2: **Overall architecture of the LiteAT system.** In Stage (a), local teachers record the classroom and themselves, which are used to reconstruct the static classroom M_c and canonical human M_h . In Stage (b), the estimated SMPL-X parameters and audio are transmitted to animate the canonical human via pose mapping. The dynamic human and classroom are combined into a complete scene M_s and rendered in the VR headset. A viewpoint recommendation module generates high-quality viewpoints through graph-based modeling and continuously refines them via a dynamic optimization mechanism.

human details. To address these limitations, Sakashita et al. [44] leverage the photorealistic, view-dependent properties of NeRF to enhance low-quality point clouds, improving visual realism. However, the long training time of NeRF-based methods makes them unsuitable for real-time human reconstruction in telepresence. Moreover, both point cloud and NeRF-based methods involve complex encoding and decoding during real-time transmission, as point clouds require large data transfer and NeRF relies on image-based streaming, resulting in high bandwidth and computational demands. Therefore, our method seeks to balance high visual quality and low latency by combining advanced rendering techniques with a lightweight data transmission strategy.

2.3 Gaussian Splatting

Gaussian Splatting is a volumetric rendering technique that represents scenes using 3D Gaussians [19], offering a compact and explicit representation that improves noise robustness and preserves geometric and color details compared to NeRF. By incorporating parameters such as time dependency and transformation information, it has been widely adopted for dynamic scene modeling [26, 29, 55] and human reconstruction [7, 13, 14, 27, 33], demonstrating strong performance in capturing motion and visual realism. Moreover, the compact nature of 3D Gaussians enables rapid optimization and real-time rendering performance. This computational efficiency supports seamless exploration of 3D scenes from arbitrary viewpoints [10], making Gaussian Splatting particularly well-suited for interactive applications such as VR environments. However, its potential in VR telepresence remains largely unexplored, particularly in remote education scenarios that demand real-time rendering and transmission of both human motion and environmental context, along with flexible and intuitive interaction within immersive VR systems.

3 LITEAT SYSTEM

We present LiteAT, a system designed to enable high-fidelity and low-latency VR telepresence for remote education. From an implementation perspective, LiteAT operates in two stages: a *Pre-Class Preparation Stage* and an *In-Class Teaching Stage*, as illustrated in Fig. 2. In the *Pre-Class Preparation Stage*, we employ a Gaussian Splatting-based reconstruction pipeline to model the entire teaching scene, including the static classroom model M_c and the canonical human model M_h , using recorded video captured by the local teacher. During the *In-Class Teaching Stage*, the SMPL-X parameters of the local teacher are

estimated and transmitted to the remote side along with the audio stream. These parameters are used to animate the canonical human model through pose mapping. The static classroom and the dynamic human are then integrated into a complete scene model M_s , which is rendered within the student's VR headset. To enhance the viewing experience, we introduce a user-adaptive viewpoint recommendation framework that supports efficient and personalized exploration. Candidate viewpoints are represented using a graph-based structure, assigning each a visual quality score. These viewpoints are continuously updated through a dynamic optimization mechanism that adjusts to recent user behaviors and scene dynamics. The following sections detail the two stages.

4 PRE-CLASS PREPARATION STAGE

In this stage, the local teacher records the classroom and themselves, which are transmitted to the remote student for reconstruction prior to the class. To support this process, we employ a Gaussian Splatting-based pipeline to independently reconstruct the static classroom and the canonical human model from the recorded video.

4.1 Preliminary

SMPL-X Model SMPL-X [37] is a unified body model with shape parameters trained jointly for the face, hands, and body, which employs standard vertex-based linear blend skinning with learned corrective blend shapes. It is defined by a function $M(\theta, \beta, \psi) : \mathbb{R}^{|\theta|} \times \mathbb{R}^{|\beta|} \times \mathbb{R}^{|\psi|} \rightarrow \mathbb{R}^3$, where θ is the whole body pose parameters, β the shape parameters and ψ the facial expression parameters. The pose can be divided into the global pose θ_g , head pose θ_f , articulated hand poses θ_h , and remaining body poses θ_b .

3D Gaussian Splatting 3D Gaussian Splatting [19] is an explicit point-based 3D representation that enables high-quality real-time rendering. This representation consists of a set of 3D Gaussians, each parameterized by mean position p , and covariance matrix Σ :

$$G(p) = e^{-\frac{1}{2}(x-p)^T \Sigma^{-1}(x-p)}. \quad (1)$$

For differentiable optimization, the covariance matrix Σ is decomposed into scaling $S \in \mathbb{R}^3$ and rotation matrix $R \in SO(3)$, such that $\Sigma = RSS^T R^T$. With these properties, rendered images can be generated using differentiable splatting [57]. Given a viewing transformation matrix W , the Jacobian matrix J of the affine approximation of the projective transformation, the covariance matrix in camera coordinates Σ' can be computed as $\Sigma' = JW\Sigma W^T J^T$. The final color is determined through the alpha blending of N sorted 3D Gaussians.

4.2 Gaussian Splatting-Based Reconstruction

4.2.1 Data Preprocessing

The recorded video covers the overall classroom environment and multi-view observations of the teacher. To separate training data for the classroom and the human, we extract video frames and apply YOLOv8 to detect the presence of a person in each frame. To ensure data quality, we discard blurry frames caused by rapid camera movement or human motion. Laplacian variance analysis [2] is used, where lower values indicate higher blurriness. Frames below an empirically defined threshold are excluded from processing.

4.2.2 Static Classroom Modeling

We reconstruct the static classroom model M_c based on the standard Gaussian Splatting approach [19], which initializes 3D Gaussians from a sparse point cloud of the scene. To obtain this point cloud, we first apply Scale-Invariant Feature Transform (SIFT) to extract key feature points and descriptors from the input images [28]. These features are matched across multiple views, and Structure from Motion (SfM) [45] is employed to generate a sparse 3D reconstruction, including camera intrinsics, extrinsics, and the 3D positions and colors of the point cloud. To enable accurate reconstruction of classroom geometry and appearance, we optimize the Gaussian parameters of the classroom model M_c by minimizing a loss function $\mathcal{L}_{\text{scene}}$ that compute pixel-wise similarity $\mathcal{L}_1$ and structural similarity $\mathcal{L}_{\text{SSIM}}$ between the rendered

image I and the ground-truth input image I_{gt} . Specifically, we define the following loss:

$$\mathcal{L}_{scene} = \lambda_1 \mathcal{L}_1 + \lambda_{SSIM} \mathcal{L}_{SSIM}, \quad (2)$$

where $\mathcal{L}_1$ denotes the L1 loss between I and I_{gt} , enforcing pixel-level accuracy by directly penalizing intensity differences, ensuring the rendered image aligns closely with the ground truth in color and detail. $\mathcal{L}_{SSIM}$ represents the Structural Similarity (SSIM) loss between I and I_{gt} , emphasizing structural information such as edges and textures. λ_1 , λ_{SSIM} control the contribution of each loss term.

4.2.3 Canonical Human Modeling

Inspired by GaussianAvatar [13], we construct a canonical human model M_h using animatable 3D Gaussians anchored to the canonical SMPL-X mesh. Each Gaussian is associated with learnable attributes, including spatial offsets, RGB color, and scale, allowing for pose-conditioned appearance variation and smooth deformation under different motion states. To obtain clean foreground data for training, we apply a keypoint-guided human segmentation strategy. We first estimate 2D keypoints using OpenPose [5] to capture the full-body skeleton, which is then passed to SAM2 [41] to produce an initial segmentation mask. The mask is refined using binary thresholding and morphological operations, followed by connected component analysis to retain the largest segment and isolate the human region. SMPL-X parameters, including body shape, pose, global orientation, and translation, are estimated for each training frame using PyMAF-X [60]. These parameters are used to deform the canonical Gaussians into pose space via linear blend skinning, enabling motion-conditioned representation and rendering. To supervise the learning of the dynamic human model, we adopt a multi-term loss function that optimizes appearance fidelity and geometric consistency by incorporating image-level reconstruction, perceptual similarity, and regularization over Gaussian parameters, formulated as:

$$\begin{aligned} \mathcal{L}_{human} = & \lambda_1 \mathcal{L}_1 + \lambda_{SSIM} \mathcal{L}_{SSIM} + \lambda_{LPIS} \mathcal{L}_{LPIS} \\ & + \lambda_f \mathcal{L}_f + \lambda_{Offset} \mathcal{L}_{Offset} + \lambda_{Scale} \mathcal{L}_{Scale}, \end{aligned} \quad (3)$$

where $\mathcal{L}_1$ is the L1 loss, encouraging pixel-level accuracy. $\mathcal{L}_{SSIM}$ is the SSIM loss, promoting structural consistency. $\mathcal{L}_{LPIS}$ [61] measures perceptual similarity in deep feature space, enhancing overall visual fidelity. $\mathcal{L}_f$ is the L2 norm of the intermediate pose-conditioned feature map, which guides the per-frame generation of appearance attributes (e.g., color and opacity), thereby improving generalization and preventing overfitting. $\mathcal{L}_{Offset}$ and $\mathcal{L}_{Scale}$ are L2 regularization terms applied to constrain the learnable Gaussian offsets and scale parameters, respectively, promoting geometric stability and preventing degenerate solutions. Each term is weighted by its corresponding coefficient λ , which balances its contribution to the overall training objective.

5 IN-CLASS TEACHING STAGE

5.1 SMPL-X Estimation

During the *In-Class Teaching Stage*, we estimate the SMPL-X parameters of the teacher from monocular RGB frames using PyMAF-X [60]. The estimated parameters, including body shape, pose, global orientation, and translation, provide a compact representation of human motion. Instead of transmitting full video or rendered images ($> 10^2$ KB), we adopt a lightweight transmission strategy that sends only the SMPL-X parameter files (approximately 2 KB per frame) along with the audio stream. This approach substantially reduces communication overhead while retaining sufficient motion detail to support accurate pose-driven reconstruction on the remote side.

5.2 Human-Classroom Integration

5.2.1 Pose Mapping

On the remote side, the received SMPL-X parameters are applied to the canonical human model M_h through pose mapping to update the teacher's motion in real time. Each 3D Gaussian in the canonical model is anchored to the SMPL-X mesh and deformed into pose space via linear blend skinning driven by the estimated parameters. This

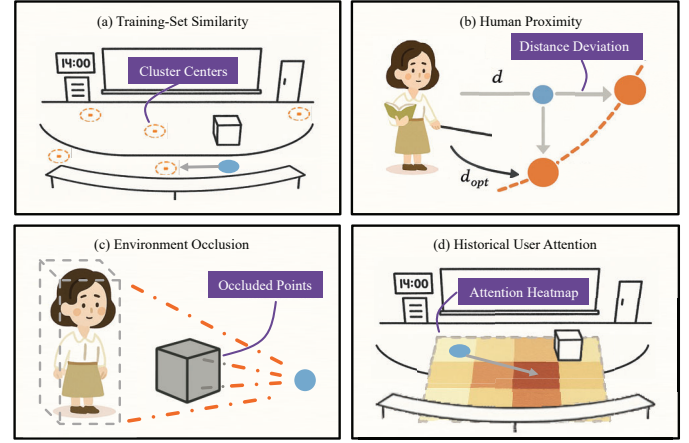


Fig. 3: **Four scoring factors for candidate viewpoints (blue point).**

(a) Training-Set Similarity based on the minimum distance to the cluster centers of training views, (b) Human Proximity measured by the deviation from an optimal viewing distance, (c) Environment Occlusion evaluated by the proportion of visible sample points on the human's bounding box, and (d) Historical User Attention computed from a voxel-based heatmap of accumulated user dwell time.

pose-conditioned transformation produces a dynamic human model that accurately reflects the teacher's current motion. Pose mapping is efficient, owing to the compactness of SMPL-X representation and the structure-preserving nature of Gaussian deformation, making it well-suited for subsequent real-time rendering.

5.2.2 Scene Combination

After obtaining the reconstructed classroom and the human model, we integrate them into a unified scene M_s for subsequent rendering. We adopt a shared camera coordinate system to ensure spatial alignment between the two components. In particular, the intrinsic and extrinsic camera parameters used for the human model are directly inherited from the sparse reconstruction of the static classroom. Since PyMAF-X estimates SMPL-X parameters under a weak perspective assumption, slight scale discrepancies may arise between the human model's predicted and actual physical dimensions. To compensate for this, we apply minor scale adjustments to the human model to improve geometric alignment. These differences are negligible and have no noticeable impact on the overall scene quality.

5.3 User-Adaptive Viewpoint Recommendation

To simplify user control and enable personalized scene exploration in LiteAT, we propose a user-adaptive viewpoint recommendation framework that balances visual quality with user preferences. Candidate viewpoints are represented as graph nodes, each scored based on four key factors that assess their visual quality. Edges encode spatial continuity to ensure smooth transitions between viewpoints. The graph is updated through a dynamic optimization mechanism that reflects recent user behavior and scene dynamics. Personalization is achieved by adjusting scoring weights and refining viewpoint positions in real-time to better align with user preferences.

5.3.1 Graph-Based Modeling

We define an undirected graph $G = (V, E)$ for viewpoint recommendation, where V comprises two types of nodes: V_{c_i} , representing the i -th candidate viewpoints, and V_u , representing the user. The set E denotes the edges connecting these nodes. Each candidate node V_{c_i} stores its spatial position P_{c_i} and a score $S(V_{c_i})$ reflecting its overall quality, while each user node stores the user's current position P_u . To evaluate the quality of candidate viewpoints, we construct a scoring function that assesses each candidate viewpoint based on four key factors, including training-set similarity S_{train} , human proximity S_{hum} , environment occlusion S_{occ} , and historical user attention S_{att} , as shown in Fig. 3. The

scoring function is shown in the following:

$$S(V_{c_i}) = w_{\text{train}}S_{\text{train}} + w_{\text{hum}}S_{\text{hum}} + w_{\text{occ}}S_{\text{occ}} + w_{\text{att}}S_{\text{att}}, \quad (4)$$

where $S(V_{c_i})$ is the overall score of candidate viewpoint V_{c_i} . The weights w_{train} , w_{hum} , w_{occ} , and w_{att} control the relative contribution of each factor to the final score, and are initially set to 0.25 to assign equal importance during the modeling stage.

Training-Set Similarity Training-set similarity score is designed to estimate the potential rendering quality of a candidate viewpoint based on its proximity to the distribution of training views, as shown in Fig. 3a. This design is motivated by the characteristics of Gaussian Splatting, which relies heavily on training views during the learning process and typically produces the highest rendering fidelity in regions covered by or near these views [63]. When the spatial coverage of training views is limited, the model often fails to reconstruct fine details in regions far from the original camera positions, resulting in degraded rendering performance. To quantify this effect, we perform K-Means clustering on the camera positions in the training set to obtain a set of representative cluster centers $\{C_k\}_{k=1}^K$, capturing the spatial distribution of the training views. For a given candidate viewpoint V_{c_i} , we compute the similarity score based on the minimum Euclidean distance to the cluster centers:

$$S_{\text{train}}(V_{c_i}) = \exp\left(-\min_k \|P_{c_i} - C_k\|_2\right), \quad (5)$$

where P_{c_i} is the spatial position of V_{c_i} . This formulation ensures that viewpoints closer to the training distribution receive higher scores while those farther away are penalized exponentially.

Human Proximity Human proximity score ensures that each candidate viewpoint maintains a suitable distance from the human, as presented in Fig. 3b. Gaussian Splatting may suffer from rendering artifacts or detail loss when the viewpoint is too close to the human. Conversely, if the viewpoint is too far, the rendered human may lack visual clarity. Therefore, the distance between the candidate viewpoint and the human is incorporated into the scoring metric. We extract the 3D bounding box of the human's Gaussian model and use its center as the human's reference point P_h . Given a candidate viewpoint V_{c_i} , we define the human proximity score as:

$$S_{\text{hum}}(V_{c_i}) = \exp\left(-\left(\|P_{c_i} - P_h\|_2 - d_{\text{opt}}\right)\right), \quad (6)$$

where d_{opt} denotes a predefined optimal viewing distance (3.0 meters in our implementation), selected based on empirical observations to minimize rendering distortion while preserving sufficient visual detail. This function assigns higher scores to viewpoints closer to the optimal distance while penalizing those that deviate significantly.

Environment Occlusion Environment occlusion score quantifies the extent to which the human is occluded by environmental objects from each candidate viewpoint, as seen in Fig. 3c. An ideal viewpoint should keep most of the human body clearly visible with minimal obstruction. To evaluate occlusion, we sample a set of points on the human's bounding box and assess their visibility from the candidate viewpoint. Specifically, we include the eight corner points and uniformly sample a 3×3 grid on each face, yielding a comprehensive set of representative points after merging and deduplication. For each point, a ray is cast from the candidate viewpoint toward the point, and its intersection interval with the human's bounding box is computed. A point is considered occluded if any projected points from the classroom Gaussian model fall within this interval. Environment occlusion score is defined as the proportion of sampled points that remain unoccluded by the environment:

$$S_{\text{occ}}(V_{c_i}) = 1 - \frac{N_{\text{occ}}(V_{c_i})}{N_{\text{total}}}, \quad (7)$$

where $N_{\text{occ}}(V_{c_i})$ is the number of occluded sampling points at viewpoint V_{c_i} , and N_{total} is the total number of sampled points. A higher score indicates that the human is largely unobstructed and clearly visible from the current viewpoint, while a lower score suggests significant occlusion by surrounding objects.

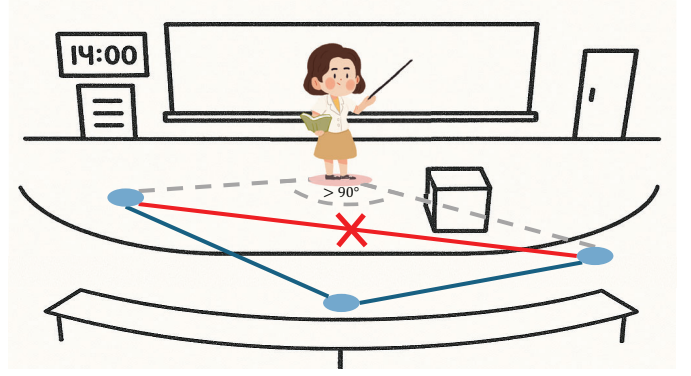


Fig. 4: **Directional continuity constraint.** An edge (blue line) is added between two viewpoints only if their angular deviation to the human is less than 90° . Otherwise, no edge (red line) is added.

Historical User Attention Historical user attention Score evaluates the long-term attention a viewpoint has received, as illustrated in Fig. 3d. During exploration, prolonged dwell time at a location often suggests that users perceive the viewpoint as offering superior quality. To model this behavior, we construct an attention heatmap H that aggregates user dwell frequencies across space. Continuous 3D space is discretized into unit-sized voxels (e.g., voxel(1.8, 3.4, 2.5) = (1, 3, 2)). Each voxel v stores a heat value $H(v)$, indicating the accumulated user dwell frequency. Upon each user position update, the corresponding voxel is determined via spatial discretization, and its heat value is incremented. Given a candidate viewpoint V_{c_i} , the historical user attention is defined as:

$$S_{\text{att}}(V_{c_i}) = \frac{H(\text{voxel}(P_{c_i}))}{H_{\text{max}}}, \quad (8)$$

where $H(\text{voxel}(P_{c_i}))$ denotes the accumulated number of visits to the voxel containing the candidate viewpoint, and H_{max} represents the highest visit count among all voxels in the scene.

Directional Continuity Constraint Once the set of nodes is determined, the corresponding set of edges E is constructed based on their pairwise relationships. To decide whether an edge $e \in E$ should be added between any two nodes, we introduce a directional continuity constraint, as shown in Fig. 4. It prevents abrupt viewpoint transitions that may cause user discomfort, such as motion sickness. Specifically, for a pair of nodes V_{c_i} and V_{c_j} , we define a directional cost function based on the angle between their direction vectors relative to the human center P_h :

$$\text{BaseCost}(V_{c_i}, V_{c_j}) = \frac{1}{\pi} \arccos\left(\frac{(P_{c_i} - P_h) \cdot (P_{c_j} - P_h)}{\|P_{c_i} - P_h\| \cdot \|P_{c_j} - P_h\|}\right). \quad (9)$$

If the value is below a predefined threshold (0.5 in our setting, equal to 90°), the transition is considered sufficiently smooth, and an edge is added between the nodes. The computed cost is then assigned as the edge weight. Otherwise, the edge is excluded.

Graph Initialization We adopt two sampling strategies to obtain reasonably good initial candidate viewpoints: clustering-based sampling and human-proximal uniform sampling. We first apply K-Means to historical camera positions from training views to extract representative viewpoints as cluster centers. To address potential sparsity near the human, we additionally sample viewpoints uniformly around the human's center on the horizontal plane, evenly distributed along a circular path with a fixed radius. These two sets are merged to form an initial pool of candidate nodes. Each node is assigned a score using the previously defined scoring function, and edges are constructed based on a directional continuity constraint to ensure smooth transitions. To reduce computational complexity, we retain the top-scoring nodes to construct the final graph and fix their number throughout the recommendation process ($N_{\text{max}} = 12$).

5.3.2 Dynamic Optimization

As the human in the scene changes in real time, we update the scores of all candidate nodes at each timestep to reflect the latest scene context. If the score of a viewpoint node falls below a predefined threshold, it is considered low quality and removed from the graph. To ensure that the number of viewpoint nodes does not fall below the predefined maximum $N_{\max}$, we supplement the graph with new candidate nodes when necessary. Specifically, we leverage an attention heatmap derived from the user's historical interaction data and identify voxels with prolonged dwell times. The centers of these voxels are transformed into world coordinates and proposed as new candidate viewpoints. Each new candidate is checked against existing nodes to ensure it satisfies a minimum distance constraint, thereby avoiding redundancy and excessive spatial density. If the constraint is met, the node is added to the graph, its score is computed using the current scoring function, and edges are established according to the directional continuity constraint. Furthermore, to maintain a high-quality set of viewpoints over time, the graph is continuously adapted by dynamically updating scoring weights and refining node positions based on the user's recent behavior and preferences. To support this process, users can freely explore the scene via controller-based translation and rotation, and their positions during navigation are recorded as behavioral signals for dynamic updates.

Scoring Weight Update We leverage recent user dwell time data to adaptively update the weights of the scoring function. A lightweight Ridge regression model [11] is used to adjust the weights of the four sub-scores, enabling the system to better reflect the user's current preferences and behaviors. To construct the training set, we label viewpoints based on dwell time: those with over 5 seconds are treated as positive samples, indicating high engagement, while those with durations between 0.5 and 2 seconds are considered negative, suggesting dissatisfaction. Samples are collected within a fixed one-minute window. Training is triggered once both positive and negative sample counts reach at least 50. If not, the window is extended, or dwell time thresholds are slightly relaxed to ensure sufficient data for learning. Given the collected samples, we construct a feature matrix X and a label vector y , where each row of X encodes the four sub-scores of a candidate viewpoint, and the corresponding element in y is the target label. The regression objective is defined as:

$$w_{\text{new}} = \arg \min_w \left\{ \|Xw - y\|^2 + \alpha \|w\|^2 \right\}, \quad (10)$$

where α is the regularization coefficient. To balance long-term user preferences with short-term behavioral fluctuations, we apply an exponential moving average to smooth weight updates over time:

$$w_{\text{updated}} = \beta w_{\text{new}} + (1 - \beta) w_{\text{old}}, \quad (11)$$

where β is the smoothing factor. This strategy mitigates abrupt changes in viewpoint scoring caused by temporary user behaviors while allowing the system to gradually adapt to evolving preferences.

Viewpoint Position Refine To further improve the quality of candidate viewpoints, we perform local refinement on each viewpoint by estimating the gradient of the scoring function using the FDM method [32]. Node scores are updated using the latest learned weights, and the estimated gradients are used to iteratively adjust the node positions. The position update of each V_{c_i} is defined as:

$$P_{c_i}(x_{\text{new}}, y_{\text{new}}, z_{\text{new}}) = P_{c_i}(x, y, z) + \alpha \nabla S(V_{c_i}), \quad (12)$$

where α is the step size, and $\nabla S(V_{c_i})$ is the gradient of the scoring function, numerically approximated via finite differences. This local refinement procedure encourages each candidate viewpoint to gradually converge toward a more optimal spatial position.

Viewpoint Recommendation In the viewpoint recommendation stage, to account for both transition smoothness and viewpoint quality, we convert the original edge weights into effective edge costs. Specifically, for each edge from node V_{c_i} to node V_{c_j} , the cost is adjusted by incorporating a penalty term:

$$C(V_{c_i}, V_{c_j}) = \text{BaseCost}(V_{c_i}, V_{c_j}) + \lambda (1 - S(V_{c_j})), \quad (13)$$

where λ is a hyperparameter that balances the trade-off between transition smoothness and viewpoint quality, and $S(V_{c_j})$ is the normalized score of node V_{c_j} . We then use Dijkstra's algorithm to compute the shortest path from the current user node V_u to each candidate viewpoint node. For a given path $P = \{V_u, V_{c_1}, V_{c_2}, \dots, V_{c_n}\}$, the cumulative path cost is computed as:

$$C(P) = \sum_{k=1}^n C(V_{c_{k-1}}, V_{c_k}), \quad \text{where } V_{c_0} = V_u. \quad (14)$$

The optimal viewpoint V_c^* is the one that minimizes the cumulative path cost from the current user node V_u :

$$V_c^* = \arg \min_{V_{c_j} \in V_c} C(P_{u \rightarrow j}), \quad (15)$$

where $P_{u \rightarrow j}$ denotes the shortest path from V_u to V_{c_j} . This approach enables the recommendation module to integrate both the graph connectivity constraint and the dynamic evaluation of viewpoint quality, resulting in more accurate and robust viewpoint selection.

6 IMPLEMENTATION

On the teacher's side, a laptop equipped with an Intel(R) Core(TM) i7-10750H CPU and an Nvidia GeForce RTX 2060 GPU is deployed, along with a microphone for capturing the teacher's voice. A Logitech C930c camera records the classroom environment and the teacher. An SMPL-X estimation module in the laptop is integrated to estimate SMPL-X parameters continuously. On the student's side, all modules run on a Windows desktop with an Intel i9-12900KF CPU and RTX 4080 GPU. Python-based computational modules handle rendering and viewpoint recommendation. Unity 2022.3 LTS is used for VR environment construction and interaction, deployed on Meta Quest Pro via Oculus Integration and OpenXR. A shared memory-based communication mechanism enables data exchange between Unity and Python. Data transmission between the local teacher and remote student sides is conducted via TCP/IP. Full implementation and demonstration videos will be publicly available at: <https://sites.google.com/view/lite-adaptive-telepresence>.

7 EVALUATION

To evaluate the effectiveness of the LiteAT system in supporting immersive remote VR education, we conducted both quantitative evaluations and user studies. The quantitative evaluation compares the performance of our method with point cloud and NeRF-based approaches. We also analyze the latency introduced by each system module to assess its suitability for VR telepresence applications. In parallel, user studies were conducted to assess the effectiveness of two core components of our system: the Gaussian Splatting-based reconstruction pipeline and the viewpoint recommendation module.

7.1 System Evaluation

To quantitatively assess the efficiency of our system, we compare it with two common baselines: the point cloud and the NeRF-based method. We conducted 50 trials for each method and calculated the average latency, with each trial covering a complete data presentation cycle. Acquisition latency is the time to capture and preprocess data, transmission is the time to send it over a local network, reconstruction is the time to reconstruct and render the scene, and total latency is their sum. All transmissions were conducted over a standard TCP/IP connection within a controlled local network environment to ensure a fair and consistent comparison. No compression or acceleration techniques were applied, allowing for an unbiased assessment of how data volume and representation format affect system latency. The three approaches are detailed in the following:

(1) Point Cloud-Based Method: A Kinect V2 captures dense RGB-D images to generate 3D point clouds, which are then streamed to the remote side for rendering, similar to LiveScan3D [21].

Table 1: Latency comparison across three telepresence approaches, including acquisition, transmission, reconstruction time, and total latency.

Method	Acquisition	Transmission	Reconstruction	Total
Point Cloud-Based Method	≈ 62.5 ms	≈ 260.9 ms	≈ 89.3 ms	≈ 412.7 ms
NeRF-Based Method	≈ 17.5 ms	≈ 48.5 ms	≈ 4934.0 ms	≈ 5000.0 ms
LiteAT (Pose Transfer)	≈ 106.5 ms	≈ 7.9 ms	≈ 70.2 ms	≈ 184.6 ms

(2) **NeRF-Based Method:** RGB sequences captured with a moving Logitech C930 camera are sent to the remote side, where a NeRF model is trained for novel view synthesis, as in SharedNeRF [44].

(3) **LiteAT (Pose Transfer):** Monocular RGB images from a Logitech C930 camera are used to estimate SMPL-X parameters, which are transmitted to the remote side for dynamic scene reconstruction and rendering via Gaussian Splatting.

As shown in Tab. 1, our method exhibits the lowest total latency (184.6 ms), significantly outperforming point cloud-based (412.7 ms) and NeRF-based (5000.0 ms) methods. This efficiency stems from using SMPL-X pose parameters to represent human motion, which requires transmitting only a small set of values (~ 2 KB). While our method introduces slightly higher acquisition latency (106.5 ms) due to SMPL-X parameter estimation, it significantly reduces transmission time (7.9 ms), ensuring superior end-to-end responsiveness. The pose parameters also enable fast model updates with low reconstruction latency (70.2 ms). In contrast, the NeRF-based method suffers from extremely high reconstruction latency (4934.0 ms) due to the heavy computational load of neural rendering, making it unsuitable for real-time telepresence. The point cloud-based method also introduces substantial transmission (260.9 ms) and reconstruction delays (89.3 ms), making it less suitable for bandwidth-constrained environments.

To gain a fine-grained understanding of the latency distribution within our proposed system, we decompose the process into five key components and measure the latency introduced by each: (1) **SMPL-X Estimation** infers SMPL-X parameters from monocular RGB input on the local side; (2) **Pose Transfer** transmits the estimated SMPL-X parameters over a TCP/IP connection; (3) **Gaussian Optimization** dynamically adapts the Gaussian-based human model to the received pose on the remote side; (4) **Binocular Rendering** synthesizes stereoscopic views for splatting-based rendering; and (5) **Unity Integration** handles user interaction and data management within the Unity engine, retrieves the stereo camera parameters, and updates the textures of the left and right eye views in the VR headset.

The results demonstrate that most components exhibit low latency, highlighting the effectiveness of our design for VR telepresence. SMPL-X parameter transmission incurs minimal overhead, with pose transfer taking only 7.9 ms. Gaussian Optimization (35.8 ms) and Binocular Rendering (23.2 ms) perform efficiently, benefiting from SMPL-driven deformation and splatting-based rendering. Unity Integration adds only moderate latency (11.2 ms), ensuring smooth synchronization with the VR display. All components remain within acceptable thresholds for real-time VR interaction. The primary latency bottleneck lies in the SMPL-X estimation module, which takes 106.5 ms to process monocular RGB input using PyMAF-X [60]. Despite recent progress in regression-based 3D pose estimation [42, 60], these methods remain computationally intensive due to their complex neural architectures. Future work may mitigate this limitation by adopting more lightweight regression models or employing hardware acceleration techniques, such as GPU-optimized inference engines or TensorRT-based deployment.

7.2 User Study

To evaluate the effectiveness of the LiteAT system from a user-centered perspective, we conducted two comparative user studies focusing on its core components: the Gaussian Splatting-based reconstruction pipeline and the integrated viewpoint recommendation module. (1) The first study compared three conditions, including 2D videoconferencing, 3D point cloud-based method, and our Gaussian Splatting-based method, evaluated by participants across five key dimensions: spatial awareness, social presence, system usability, visual quality, and perceived real-

time responsiveness. (2) The second study examined the impact of the viewpoint recommendation module by comparing two versions of our system: one with free navigation and the other with viewpoint recommendation. Participants rated their experiences on cognitive load reduction, controllability, preference alignment, and overall quality.

Participants We recruit 26 participants from our university and randomly assign them to two groups of 13. The first group, which completes the reconstruction pipeline evaluation, consists of 9 male and 4 female participants aged 21–28 years ($M = 23.6$, $SD = 1.85$). The second group, which completes the viewpoint recommendation module evaluation, includes 8 male and 5 female participants aged 21–25 years ($M = 23.2$, $SD = 1.30$). All participants are undergraduate or graduate students with normal or corrected-to-normal vision, and none report any visual or vestibular disorders (*e.g.*, color blindness or dyschromatopsia). All participants have prior experience using videoconferencing tools such as Zoom and Microsoft Teams. To mitigate potential confounding factors such as motion sickness or unfamiliarity with the interface, we ensure that all participants have previous experience with VR headsets and are familiar with the controllers. Each participant receives a compensation of \$50.

Study Setup A within-subject design was employed in each of the two user studies to independently evaluate the effectiveness of the reconstruction pipeline and the viewpoint recommendation module. The experiments were conducted in a controlled indoor environment, with participants wearing Meta Quest Pro headsets. All conditions were implemented using a custom Unity application. One of the authors served as the instructor and delivered a brief lesson on a classical Chinese poem. To simulate remote teaching, the instructor and participants were situated in separate rooms. To ensure clear delivery of instructional content, the teacher's voice was recorded using an external microphone and transmitted to the built-in speakers of the participants' VR headsets. This audio configuration was uniformly applied across all conditions to maintain consistency.

7.2.1 Reconstruction Pipeline Evaluation

In this experiment, we evaluate the design of the Gaussian Splatting-based reconstruction pipeline. We compare three methods:

C1. 2D Videoconferencing: This condition simulates a 2D videoconferencing setup, where a Logitech C930 camera captures the teacher and streams the video to a virtual screen in the VR headset.

C2. 3D Point Cloud-based Method: The teacher and classroom are captured with a Microsoft Kinect V2 to generate 3D point clouds, which are transmitted to the participant's VR headset for rendering.

C3. LiteAT (Gaussian Splatting-based Method): This condition uses our proposed Gaussian Splatting-based reconstruction pipeline to render the teacher and the classroom within the VR headset.

In all three conditions, participants navigated freely using the joystick input of the VR controller within the environment without assistance from the viewpoint recommendation module.

Procedures Before the experiment, participants sign an informed consent form and receive a detailed briefing on the experimental procedures. Participants experienced three conditions (C1, C2, C3) presented in randomized order, with each condition consisting of an 8-minute instructional session followed by a 2-minute break. After the sessions, participants complete a questionnaire that evaluates five metrics for the three conditions: spatial awareness, social presence, system usability, visual quality, and perceived real-time responsiveness. We adopt a five-point Likert scale, where a score of one means "strongly disagree" and a score of five means "strongly agree." Finally, after completing

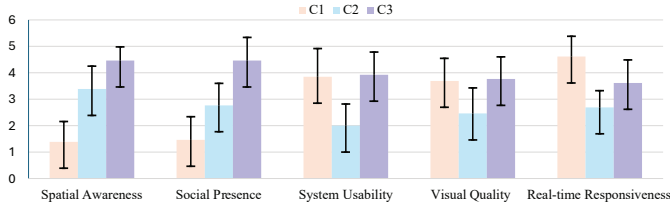


Fig. 5: **Comparison between C3 and baselines (C1, C2) based on human evaluation.** We show spatial awareness, social presence, system usability, visual quality, and real-time responsiveness evaluation scores. Error bars denote the standard errors of the mean.

all post-study questionnaires, we conduct a debriefing interview with open-ended questions.

Results We conduct repeated-measures ANOVA tests to evaluate the effect of the three methods on each metric. The means and standard deviations for the five metrics are presented in Fig. 5.

Spatial Awareness. This evaluation assesses how effectively the system supports users' perception and understanding of spatial relationships in the environment. The C3 method received the highest rating ($M = 4.46$, $SD = 0.52$), followed by C2 ($M = 3.38$, $SD = 0.87$) and C1 ($M = 1.38$, $SD = 0.77$). Mauchly's test indicated that the assumption of sphericity was met ($\chi^2(2) = 2.28$, $p = .320$). The effect of the three methods on spatial awareness score was significant at the .05 level ($F_{[2,24]} = 81.95$, $p < .001$, $\eta^2 = .87$). Post-hoc pairwise comparisons with a Bonferroni adjustment indicated that the spatial awareness score was significantly higher for the C3 method compared to both C1 ($p < .001$) and C2 ($p = .008$). Similarly, the score for C2 was significantly higher than that for C1 ($p < .001$).

Social Presence. Social presence measures the extent to which the system fosters a sense of physical presence and social engagement. The C3 method received the highest rating ($M = 4.46$, $SD = 0.88$), followed by C2 ($M = 2.77$, $SD = 0.83$) and C1 ($M = 1.46$, $SD = 0.88$). Mauchly's test showed that the assumption of sphericity had been violated ($\chi^2(2) = 8.19$, $p = .017$), and therefore degrees of freedom were corrected using Greenhouse-Geisser estimates of sphericity ($\epsilon = 0.66$). The effect of the three methods on social presence score was significant at the .05 level ($F_{[1.312,15.738]} = 44.54$, $p < .001$, $\eta^2 = .79$). Post-hoc pairwise comparisons with a Bonferroni adjustment showed that the social presence score was significantly higher at the C3 method than at both C1 ($p < .001$) and C2 ($p < .001$). C2 also scored significantly higher than C1 ($p = .008$).

System Usability. This evaluation assesses the system's ease of use, intuitiveness, and efficiency in supporting remote education. The C3 method received the highest rating ($M = 3.92$, $SD = 0.86$), slightly outperforming C1 ($M = 3.85$, $SD = 1.07$), while C2 scored the lowest ($M = 2.00$, $SD = 0.82$). Mauchly's test indicated that the assumption of sphericity had been met ($\chi^2(2) = 5.50$, $p = .064$). The effect of the three methods on system usability score was significant at the .05 level ($F_{[2,24]} = 20.72$, $p < .001$, $\eta^2 = .63$). Post-hoc pairwise comparisons with a Bonferroni adjustment indicated that the system usability score was significantly higher for the C3 method than for C2 ($p < .001$). Similarly, C1 scored significantly higher than C2 ($p < .001$). However, there was no significant difference between C3 and C1 in system usability scores ($p = 1.000$).

Visual Quality. Visual quality aims to evaluate the clarity, realism, and aesthetic fidelity of the system's visual output. The C3 method received the highest rating ($M = 3.77$, $SD = 0.83$), followed by C1 ($M = 3.69$, $SD = 0.85$) and C2 ($M = 2.46$, $SD = 0.97$). Mauchly's test indicated that the assumption of sphericity had been met ($\chi^2(2) = 0.440$, $p = .802$). The effect of the three methods on visual quality score was significant at the .05 level ($F_{[2,24]} = 9.33$, $p = .001$, $\eta^2 = .44$). Post-hoc pairwise comparisons with a Bonferroni adjustment showed that both C3 ($p = .003$) and C1 ($p = .011$) scored significantly higher than C2. However, there was no significant difference between C3 and C1 ($p = 1.000$).

Real-time Responsiveness. This evaluation assesses how effectively the system minimizes delays in visual updates and user interactions. C1 received the highest rating ($M = 4.62$, $SD = 0.77$), followed by C3 ($M = 3.62$, $SD = 0.87$) and C2 ($M = 2.69$, $SD = 0.63$). Mauchly's test showed that the assumption of sphericity was violated ($\chi^2(2) = 7.19$, $p = .027$), and therefore degrees of freedom were corrected using Greenhouse-Geisser estimates of sphericity ($\epsilon = 0.68$). The effect of the three methods on real-time responsiveness score was significant at the .05 level ($F_{[1.351,16.216]} = 19.75$, $p < .001$, $\eta^2 = .62$). Post-hoc pairwise comparisons with a Bonferroni adjustment revealed that C3 significantly outperformed C2 ($p = .048$), and C1 significantly outperformed C2 ($p < .001$). However, there was no significant difference between C1 and C3 ($p = .062$).

7.2.2 Viewpoint Recommendation Module Evaluation

To evaluate the viewpoint recommendation module, we compare two methods that rely on the same Gaussian Splatting-based reconstruction pipeline for rendering but differ in movement mode.

D1. Free Navigation: Participants navigate freely using joystick input without any system-generated guidance.

D2. LiteAT (Viewpoint Recommendation): Participants receive a set of candidate viewpoints and select their preferred one for navigation, guided by our proposed viewpoint recommendation module.

Procedures Prior to the experiment, participants received procedural instructions and provided informed consent. Each participant then completed two conditions (D1, D2) presented in randomized order, followed by four Likert-scale questions evaluating the effectiveness of the viewpoint recommendation module (D2) relative to free navigation (D1). These questions assessed cognitive load reduction, controllability, preference alignment, and overall viewing quality, with a score of 5 indicating a strong preference for D2, 1 for D1, and 3 representing neutrality. An open-ended debriefing interview was conducted at the end of the session to collect qualitative feedback.

Results We conducted Wilcoxon signed-rank tests to examine whether participants' ratings significantly differed from the neutral baseline score of 3 to evaluate the relative effectiveness of the two methods. The mean ratings and standard deviations are shown in Fig. 6. The detailed results are as follows: **(1) Cognitive Load Reduction**, which assesses how effectively the interface reduces mental effort during navigation, received an average rating of 4.54 ($SD = 0.66$). Wilcoxon signed-rank test indicated a significant difference from the neutral score ($Z = -3.176$, $p = .001$). **(2) Controllability**, reflecting users' perceived sense of control over their viewing experience, was rated at 4.08 ($SD = 0.49$). The difference was statistically significant ($Z = -3.276$, $p = .001$). **(3) Preference Alignment**, which assesses the extent to which the system aligns with users' preferred views, received a mean rating of 4.23 ($SD = 0.60$), significantly exceeding the neutral baseline ($Z = -3.176$, $p = .001$). **(4) Overall Quality**, representing participants' general assessment of the recommended viewpoints, achieved a mean score of 4.62 ($SD = 0.65$), with the difference being significant ($Z = -3.217$, $p = .001$). These results indicate that the viewpoint recommendation module (D2) performs significantly better than the free navigation method (D1) across all assessed dimensions.

8 DISCUSSION

The main feature of this research lies in adopting a Gaussian Splatting-based reconstruction pipeline with lightweight data transmission, enabling low-latency, high-fidelity remote learning experiences. Also, the integrated viewpoint recommendation module provides a more flexible and efficient means of navigating the virtual classroom. These contributions offer a new perspective on immersive remote VR education, paving the way for future advancements in real-time, interactive, and collaborative learning environments.

8.1 Key Findings

Through quantitative evaluations and user studies, our method demonstrated lower latency than baseline approaches and achieved real-time responsiveness comparable to 2D videoconferencing. This performance

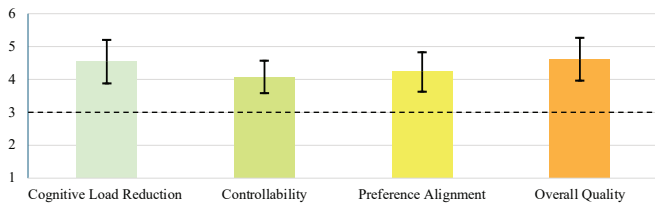


Fig. 6: **Results of the viewpoint recommendation module.** User ratings are presented for cognitive load reduction, controllability, preference alignment, and overall quality. Error bars denote the standard errors of the mean. The dashed line indicates the neutral baseline, with higher scores favoring D2 and lower scores favoring D1.

is attributed to the use of lightweight pose parameter transmission instead of dense 3D data or image sequences, combined with the real-time rendering capabilities of Gaussian Splatting. The user studies further confirmed that our system enhances spatial awareness and social presence. Notably, the 3D methods consistently outperformed the 2D baseline on both metrics, reinforcing that the spatial depth and interactive freedom provided by immersive environments support more effective perception and communication in remote teaching contexts [40, 49]. As noted by participants during the post-study interview, our system improved their understanding of the environment and enabled a clearer perception of the teacher's nonverbal cues. Gaze and gesture are known to play critical roles in expressing intention and guiding attention [46]. As one participant remarked, "*seeing the teacher's gestures from different angles helped me better follow the instruction*," and another noted, "*the sense of being in the same room made it easier to concentrate*."

Our method demonstrates comparable system usability to 2D video-conferencing, suggesting that the added spatial exploration and interactivity in virtual environments do not compromise ease of use. This was echoed by several participants, one of whom commented that "*even though it's 3D, it didn't feel harder to use than Zoom*," highlighting the intuitive design of our system. Furthermore, the photorealistic rendering capabilities of Gaussian Splatting enable our approach to match 2D video in visual quality. Participants frequently praised the visual realism, with remarks such as "*the lighting and reflections made the scene feel natural*" and "*I could clearly see facial details and gestures*." Although the point cloud-based method and our approach employ point-based representations, the latter's compact and anisotropic formulation more effectively captures fine structural details, including surface textures, reflections, and subtle variations. These insightful opinions suggest the potential of Gaussian Splatting technology in more immersive, responsive, and socially rich remote teaching environments.

A flexible and switchable navigation mode in VR environments is also essential for preserving user autonomy, as emphasized by several participants. For example, one participant shared that "*having control over where to look made me feel more involved and less like a passive viewer*," highlighting the necessity of our viewpoint recommendation module in supporting effective remote VR education. Several participants also reported heightened emotional connection and prolonged attention in the virtual classrooms. Comments such as "*I felt more emotionally connected to the teacher*" and "*time seemed to pass faster in the VR session*" suggest that immersive VR environments can enhance both cognitive and emotional engagement in remote learning.

8.2 Broader Applications

Building on the findings above, LiteAT exhibits strong potential for broader applications that demand real-time performance, photorealistic rendering, rich interaction, or enhanced social presence.

Laboratory Training. Laboratory training typically requires high spatial fidelity and responsive interaction, making it well-suited for LiteAT. For example, when students operate an electrophoresis device, the system enables detailed exploration of the 3D equipment layout and real-time observation of the instructor's hand movements. This improves understanding of complex setups and allows integration with sensor feedback for contextual guidance or safety alerts.

Procedural Teaching. Procedural teaching, such as surgical training or engineering workflows, frequently involves fine motor actions that are difficult to convey through 2D representations. LiteAT's photorealistic rendering and real-time pose transfer enable more precise and nuanced demonstrations of physical operations and can be further enhanced with wearable devices or AI-generated procedural cues to support interactive rehearsal and timely feedback.

Collaborative Learning. In collaborative learning contexts where multi-user coordination and shared attention are essential, LiteAT shows strong potential to effectively support synchronized spatial understanding and foster a strong sense of co-presence. Its compatibility with co-annotation tools and shared virtual workspaces further suggests its promise for enabling joint exploration, peer discussion, and co-creation in immersive educational settings.

8.3 Limitation and Future Works

Despite promising results and broader implications, LiteAT has four limitations that highlight valuable directions for future research.

System Latency. While the system benefits from lightweight data transmission, noticeable latency remains due to the computational cost of pose estimation. Maintaining high frame rates (e.g., 90–120 Hz) is critical for comfortable VR telepresence. Future work will explore hardware acceleration and model compression strategies.

Human-Object Interaction. While the system supports dynamic human reconstruction via Gaussian Splatting, it lacks modeling of human-object interactions, limiting its effectiveness in hands-on scenarios such as device assembly and scientific experiments. We envision enhancing it with physical object modeling and interaction tracking to support task-oriented, interactive learning environments.

Multimodal Communication. Our system provides immersive visuals and voice communication, but lacks multimodal cues such as gaze and pointing to convey shared attention. This limits the expression of subtle instructional intent and effective coordination. Future work will integrate these cues to facilitate clearer guidance and improve communicative effectiveness in virtual classrooms.

Peer Presence. The lack of peer representation limits social interaction and diminishes the sense of collaborative learning. Users may perceive the environment as isolated and teacher-centered. Future work will introduce multi-user participation and positioning [22] to enhance group presence and support a more engaging, collaborative experience.

9 CONCLUSION

In this paper, we presented *LiteAT*, a data-lightweight and user-adaptive VR telepresence system that supports real-time, high-fidelity remote learning experiences. The system integrates a Gaussian Splatting-based reconstruction pipeline to model the teacher and classroom environment and a viewpoint recommendation module that suggests optimal viewpoints. A dynamic human model driven by SMPL-X supports lightweight data transmission, while both the animated human and static classroom are jointly rendered in the students' VR headsets. The user-adaptive viewpoint recommendation module employs a graph-based representation of candidate viewpoints, incorporating multiple visual quality factors and dynamically adapting to users' recent preferences and changes in the scene. We conducted quantitative evaluations and user studies to validate the effectiveness of LiteAT. Results demonstrate that the Gaussian Splatting-based reconstruction pipeline significantly reduces latency, improving spatial awareness, social presence, and visual fidelity. The viewpoint recommendation module also enhances adaptability, control, and overall viewing quality. Our system paves the way for future advancements in versatile and scalable telepresence, such as procedural teaching and collaborative learning.

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